1) First Derivative: Rate of Reaction

The first derivative shows show the concentration of a reactant or product changes over time. This is most common derivative in chemical kinetics.

Ex: For a reation $A \to B$, the concentration of A decreases over time.

The reation is: $\text{Rate} = -\frac{d[A]}{dt}$

If the reation is first-order, the rate law is:

$$\text{Rate} = K[A]$$

The concentration of A at time t can be written as:

$[A] = [A_0]_0 e^{-Kt}$ Here, $[A]_0$ = Initial concentration A

1st derivative rate: $\frac{-d[A]}{dt} = K[A]$

2) Second Derivative: Change in Rate (Acceleration)

The second derivative shows how the rate of reaction changes over time. It's the derivative of the rate.

Ex: Second Derivative in Concentration Change:

For a first-order reaction $A \to B$:

From this equation: $\frac{d[A]}{dt} = -K[A]_0 e^{-Kt}$

The second derivative with respect of time is:

$$\frac{d^2[A]}{dt^2} = K^2[A]_0 e^{-Kt}$$

Meaning: The second derivative shows that the rate is decreasing over time (Negative slope) because A is consumed. The curvature indicates the acceleration of the rate decrease.

3) Third Derivative: Rate of Change of Acceleration

The 3rd derivative is the derivative of the second derivative. It tells how the acceleration of the rate changes over time.

Ex: Third Derivative in Kineties

For the same first-order reaction $A \rightarrow B$,

Starting from: $\frac{d^2[A]}{dt^2} = k^2[A]_0 e^{-kt}$

The third derivative is: $\frac{d^3[A]}{dt^3} = -k^3[A]_0 e^{-kt}$

Meaning: The third derivative tells us that the rate of change of acceleration is constant for this exponential decay process.

Key Note:

(i) First derivative: Most common in reaction rate calculations.

(ii) Second derivative: Usefull of study how the rate changes over time.

(iii) Third derivative: Rare in practical chemistry but shows deeper changes in kinetics.

Gold এর e^- বিন্যাস:

Gold has 79 electors arranged in different orbitals:

It's electron configuration is:

$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^6 4f^{14} 5d^{10} 6s^1$

Bond Energy: Bond energy is defind as the energy required to split the molecule into atoms. It depends on types of bonding - single, double or triple bonds.

Chemical BonD

Defination: A chemical bond is the force of attraction that holds atoms or ~~ionos~~ ions together in a molecule or compound.

Types of Chemical Bonds:

1. Ionic Bond: The bond formed when ~~electo~~ electrons are transferred from one atom to another, creating oppositely charged ions that attract each other.
   Example: Na^+ and Cl^- from $NaCl$.
2. Covalent Bond: Formed when atoms share one or more pairs of electrons to achived stability.
   Example: $H + O_2 \rightarrow H_2O$ (water); H_2O involves shared electorns between hydrogen and oxygen.
3. Hydrogen Bond: A weak bond formed between a hydrogen atom and another electronegative atom.
   Example: The bonds between water molecules.
4. Van der Waals Forces: Weak attractions between molecules or atoms due to temporary dipoles.
5. Coordinate Covalent Bond: ~~At a type of covalent to~~ A coordinate covalent bond is a type of covalent bond in which both shared electrons in the bond comes from the same atom.
   Example: When ammonia (NH_3) reacts with a ~~pe~~ proton (H^+) to form an ammonium ion (NH_4^+): $NH_3 + H^+ \longrightarrow NH_4^+$
   The lone pair is donated to the H^+, forming a coordinate covalent bond.
6. Metallic Bond: Found in metals, where valence electrons are share in a "sea of electorons" that moves freely around metal ions.
   Example: Metallic bonding in Cu (copper)

Valence Electron: The electrons in the outer most energy level in an atom that takes part in chemical bonding are called valence electrons.

Bonding electron: The valence electrons actually involved in bond formation are called bonding electrons.

Octel Rule: The tendency for atoms to have eight electrons in the outer shell is known as Octet rule.

Dipole-Dipole Forces: A type of intermolecular force that occur between polar molecules. These forces arise due to the attration between the positive end of one polar molecule and the negative end of another polar molecule.

The dipol-dipole interaction occurs as follows:

$$H\delta^+ - Cl\delta^- \cdots \delta^+H - Cl\delta^-$$

The positive end (H) of one HCl molecule is attracted to the negative end (Cl) of another HCl molecule.

London Dispersion Forces: Are weak intermolecular forces that arise due to temporary fluctuations in the electron distribution of atoms or molecule.

Bond lenth: Bond lenth is the average distance between the nuclei of two bonded atoms in a molecule.

Bond Angle: The angle between the directions of two bonds in a molecule is called the 'bond angle'

Bond angle depends on: (i) Charge distribution (ii) Geometry of the molecules (iii) Symmetry (iv) Hybiridization

Chemistry

Characteristics of different types of Bonds.

① Ionic Bond:

(i) Strong electrostatic attraction between ions.
(ii) High melting and boiling points.
(iii) Soluble in water; conduct electricity in molten or aqueous State

Example: NaCl (Sodium chloride)

② Covalent Bond:

(i) Strong bonds within molecules.
(ii) Can be polar or non polar
(iii) Low melting and boiling points.
(iv) Poor conductors of electricity.

Example: H_2O

③ Metallic Bond:

(i) Strong bond within metals.
(ii) Good conductors of heat and electricity.
(iii) Malleable and ductile
(iv) Lustrous appearance.

Example: Cu.

④ Hydrogen Bond

① Weaker then covalent bonds but stronger than van de waals force.

② Responsible for high boiling points in substance like water

③ important in DNA structure and protine folding

Example: Between water molecules H_2O (H_2O).

⑤ Van der waals forces:

① weakest types of attraction

② increases with molecular size and surface area

③ Present in molecules but significant in nonpolar substances

Example: Attraction in nobel Gases like (Ar) (argon).

⑥ Coordinate Covalent Bond

① simmilar strength to regular covalent bonds.

② Often forms between a lone pair donor and an electron deficient acceptor.

③ Common in complex ions and ligands.

Example: $NH_3 + H^+ \rightarrow NH_4^+$

Chemical Kinetics

The branch of Physical chemistry which deals with the rare reactions is called Chemical Kinetics

Ex: $A \rightarrow B$

The rate of the reaction $= -\frac{d[A]}{dt}$ [of A]

$= +\frac{d[B]}{dt}$ [of B].

Units of rate:

mole/litre sec or $mol\, l^{-1} s$.

mole/litre ~~sec~~ min or $mol\, l^{-1} min^{-1}$.

mole/litre hour or $mol\, l^{-1} h^{-1}$.

Rate ~~Con~~: Laws:

The rate of a reaction is directly proportional to the reactant concentrations. ~~each con~~

Rate $\quad rate \propto [A]^n$

$\Rightarrow rate = K[A]^n$.

For a Reaction:

$$2A + B \longrightarrow \text{Products.}$$

$$rate = K[A]^m [B]^n$$

Moleculerity of a Reaction

moleculerity of a reaction refers to the number of moleculer atom participating in an elementary ~~stem~~ step.

Chemical reactions are two types.

① Elementery Reaction ; it occurs in a single step

② Complex reaction ; it occurs in two or more step.

Examples of Rate laws:

① $2N_2O_5 \longrightarrow 4NO_2 + O_2$. rate $= k[N_2O_5]$

② $H_2 + I_2 \longrightarrow 2HI$. rate $= k[H_2][I_2]$

③ $2NO_2 \longrightarrow 2NO + O_2$ rate $= k[NO_2]^2$.

④ $2NO + 2H_2 \longrightarrow N_2 + 2H_2O$ rate $= k[H_2][NO]^2$.

Order of Reaction:

The order of a reaction is defined as the sum of the powers of concentrations in the rate law.

$$\text{rate} = k[A]^m[B]^n$$; Order of reaction is $(m+n)$.

Rate law	Reaction order
Example: rate $= k[N_2O_5]$	1.
rate $= k[H_2][I_2]$	2.

Zero Order Reaction.

A zero order reaction is one where rate is independent of concentration.

Example: $NO_2 + CO \longrightarrow NO + CO_2$ at $200°c$ is rate $= k[NO_2]$

Order of a Reaction	Molecularity of a Reaction
① it is the sum of powers of the concentrations of the terms in the law ~~exp~~ expression	① It is number of reacting species understanding simultaneous collision in the elementary or simple reaction
② it is a experimentally determind value	② It is a theoretical concept
③ It can have fractional value	③ It is always a whole number
④ It can assume zero value	④ I can not have zero value.
⑤ Order of a reaction can change with the conditions such as pressure, temperature concentration	⑤ Molecularity is invariant for a chemical equation

Arrhenius Equations

$$K = Ae^{-E_a/RT}$$

$$\begin{bmatrix} R = 8.314\ J mol^{-1} K^{-1} \\ e \approx 2.718 \end{bmatrix}$$

Where

Logarithm form of Arrhenius Equation.

$$\ln K = \ln Ae^{-E_a/RT} = \ln A - \frac{E_a}{R} \cdot \frac{1}{T}$$

$$\log K = \frac{-E_a}{2.303\ RT} + \log A.$$

Ex1: $FeCl_2 + Cl_2 \rightarrow FeCl_3$

Cl - জারক (গ্রহণ)

$FeCl_2$ - বিজারক (ত্যাগ)

~~$FeCl_2$~~

$2Fe^{2+} - 2e^- \rightarrow 2Fe^{3+}$

$Cl_2^{0} + 2e^- \rightarrow 2Cl^-$

$2Fe^{2+} + Cl_2^{0} \rightarrow 2Fe^{3+} + 2Cl^- \Rightarrow 2Fe^{2+} + Cl_2^{0} \rightarrow 2FeCl_3$

$Cl_2^{0} + 2e \rightarrow 2Cl^-$ ——(i)

$FeCl_2 - e \rightarrow Fe^{3+} + 2Cl^-$ ——(ii)

(ii) × 2 for equal the elector number,

$Cl_2 + 2e = 2Cl^-$

$2FeCl_2 - 2e = 2Fe^{3+} + 4Cl^-$

$2FeCl_2 + Cl_2 = 2Fe^{3+} + 6Cl^-$

$\Rightarrow 2FeCl_2 + Cl_2 = 2FeCl_3$

Ex2: $Br_2 + 2e = 2Br^{-1}$ ——(i)

~~$H_2S - 2e = S + 2H^+$~~

$H_2S - 8e^- + 4H_2O = H_2SO_4 + 8H^+$ ——(ii)

(i) × 4 then add (i) + (ii)

$4Br_2 + 8e = 8Br^-$

$H_2S - 8e^- + 4H_2O = H_2SO_4 + 8H^+$

$4Br_2 + H_2S + 4H_2O = H_2SO_4 + 8HBr$

Ex3: $K_2Cr_2O_7 + 6e^- = 2Cr^{3+} + 2K^+ + 7O_2^-$ ——(i)

$2KI - 2e^- = I_2^{0} + 2K^{+1}$ ——(ii)

(ii) × 3 then add (i) + (ii)

$K_2Cr_2O_7 + 6e^- = 2Cr^{3+} + 2K^+ + 7O_2^-$

$6KI - 6e^- = 3I_2 + 6K^+$

$K_2Cr_2O_7 + 6KI = 2Cr^{3+} + 2K^+ + 7O_2^- + 3I_2 + 6K^+$

$\Rightarrow K_2Cr_2O_7 + 6KI + 14H^+ = 2Cr^{3+} + 8K^+ + 7H_2O + 3I_2$

$\Rightarrow K_2Cr_2O_7 + 6KI + 14HCl = 2CrCl_3 + 8KCl + 7H_2O + 3I_2$

$FeCl_2 + Cl_2 \rightarrow FeCl_3$

$Cl_2 + 2e \rightarrow 2Cl^-$ ——(i)

$FeCl_2 - e \rightarrow Fe^{3+} + 2Cl^-$ ——(ii)

(ii) × 2 + (i) + (ii).

$Cl_2 + 2e \rightarrow 2Cl^-$

$2FeCl_2 - 2e \rightarrow 2Fe^{3+} + 4Cl^-$

$2FeCl_2 + Cl_2 \rightarrow 6Cl^- + 2Fe^{3+}$

$\Rightarrow 2FeCl_2 + Cl_2 \rightarrow 2FeCl_3$

$Br_2 + 2e^- \rightarrow 2Br^-$

$H_2S - 8e^- + 4H_2O \rightarrow H_2SO_4 + 8H^+$

If k_1 & k_2 are the values of rate constant, at temperature T_1 & T_2, we can write,

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left[\frac{T_2 - T_1}{T_2 T_1}\right]$$

Half life of a Reaction : Half-life ($t_{1/2}$) in chemistry is the time required for half of a reactant to be consumed in a chemical reaction.

1. Zero-Order Reaction: $t_{1/2} = \frac{[A]_0}{2k}$
2. First-Order Reaction: $t_{1/2} = \frac{0.693}{k}$
3. Second-Order Reaction: $t_{1/2} = \frac{1}{k[A]_0}$
4) The intergrated rate ~~of~~ equation for a first order reaction, $k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$

$$\boxed{\cancel{\text{or } k = \frac{\ln 2}{t}}} \text{ or } k = \frac{1}{t} \ln \frac{[A]_0}{[A]}$$

half-life : $t_{1/2} = \frac{\ln 2}{k}$

5 Math: for first ~~ado~~ order reaction,

$$t_{1/2} = \frac{0.693}{k}$$

$$\Rightarrow k = \frac{0.693}{t_{1/2}} = \frac{0.693}{100}$$

$$= 0.00693\ s^{-1}$$

When $75\% = \frac{3}{4}$ ~~intial~~ initial concentration has reacted, it is reduced to $\frac{1}{4}$.

Substituting values in the rate equation

$$t_{3/4} = \frac{1}{k} \ln \frac{[A]_0}{\frac{1}{4}[A]_0}$$

$$= \frac{1}{0.00693} \ln (4)$$

$$= 200 \text{ sec (Ans:)}$$

6. $t_{\frac{1}{2}} = \frac{0.693}{K}$

$$= \frac{0.693}{0.450} = 1.54 \text{ sec}$$

জারক e^- গ্রহণ বিজারণ অর্ধ বিক্রিয়া :

$KMnO_4$, $K_2Cr_2O_7$, I_2

(1) $KMnO_4 + 5e^- + 8H^+ \rightarrow K^+ + Mn^{2+} + 4H_2O$

(2) $K_2Cr_2O_7 + 6e^- + 14H^+ \rightarrow 2K^+ + 2Cr^{3+} + 7H_2O$

(3) $I_2 + 2e^- \rightarrow 2I^-$

বিজারক : e^- ত্যাগ জারণ অর্ধবিক্রিয়া :

$FeSO_4$, $FeCl_2$, KI, H_2S, H_2O_2, $H_2C_2O_4$, $Na_2S_2O_3$

(1) $Fe^{2+} - e^- \rightarrow Fe^{3+}$

(2) $Fe^{2+} - e^- \rightarrow Fe^{3+}$

(3) $2KI - 2e^- \rightarrow 2K^+ + I_2$

(4) $H_2S - 2e^- \rightarrow S + 2H^+$

(5) $H_2O_2 - 2e^- \rightarrow 2H^+ + O_2$

(6) $H_2C_2O_4 - 2e^- \rightarrow 2H^+ + 2CO_2$

(7) $2S_2O_3^{2-} - 2e^- \rightarrow S_4O_6^{2-}$

Ex2: $4Br_2 + 8e^- \rightarrow 8Br^-$

$H_2S - 8e^- + 4H_2O \rightarrow H_2SO_4 + 8H^+$

$4Br_2 + H_2S + 4H_2O \rightarrow H_2SO_4 + 8HBr$

Ex3: $K_2Cr_2O_7 + 6e = 2K^+ + 2Cr^{3+} + 7O^{-2}$ —— (i)

$2KI - 2e = I_2 + 2K^+$ —— (ii)

(ii)×3 and (i) + (ii)

$6KI - 6e = 3I_2 + 6K^+$

$K_2Cr_2O_7 + 6e = 2K^+ + 2Cr^{3+} + 7O^{2-}$

$6KI + K_2Cr_2O_7 = 3I_2 + 8K^+ + 2Cr^{3+} + 7O^{2-}$

$\Rightarrow 6KI + K_2Cr_2O_7 + 14H^+ = 3I_2 + 8K^+ + 2Cr^{3+} + 7H_2O$

$\Rightarrow 6KI + K_2Cr_2O_7 + 14HCl = 3I_2 + 8KCl + 2CrCl_3 + 7H_2O$

Ex4: $KMnO_4 + 5e^- = Mn^{2+} + K^+ + 4O^{-2}$ —— (i)

$H_2S - 2e^- = 2H^+ + S^\circ$ —— (ii)

(i)×2 and (ii)×5; then add (i)+(ii)

$2KMnO_4 + 10e^- = 2Mn^{2+} + 2K^+ + 8O^{-2}$

$5H_2S - 10e^- = 10H^+ + 5S^\circ$

$2KMnO_4 + 5H_2S = 2Mn^{2+} + 2K^+ + 8O^{-2} + 10H^+ + 5S$

$\Rightarrow 2KMnO_4 + 5H_2S + 6H^+ = 2Mn^{2+} + 2K^+ + 8H_2O + 5S$

$\Rightarrow 2KMnO_4 + 5H_2S + 3H_2SO_4 = 2MnSO_4 + K_2SO_4 + 8H_2O + 5S$

Ex5: $HNO_3 + e = NO_2 + H^+ + O^{-2}$ —— (i)

$CuS - 2e = Cu^{2+} + S^\circ$ —— (ii)

(i)×2 and (i)+(ii),

$2HNO_3 + 2e = 2NO_2 + 2H^+ + 2O^{-2}$

$CuS - 2e = Cu^{2+} + S$

$2HNO_3 + CuS = 2NO_2 + 2H^+ + 2O^{-2} + Cu^{2+} + S$

$\Rightarrow 8HNO_3 + CuS = 8NO_2 + 4H_2O + \cancel{4H_2O} + CuSO_4$

Ex6: $I_2 + 2e^- \rightarrow 2I^-$

$2S_2O_3^{2-} - 2e^- \rightarrow \cancel{SO_4}\ S_4O_6^{2-}$

$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{-2}$

$\Rightarrow I_2 + 2Na_2S_2O_3 \rightarrow 2NaI + Na_2S_4O_6$

Ex6: $2Na_2S_2O_3 - 2e^- \rightarrow Na_2S_4O_6 + 2Na^+$

$I_2 + 2e^- \rightarrow 2I^-$

$2Na_2S_2O_3 + I_2 \rightarrow Na_2S_4O_6 + 2NaI$

Ex: $KMnO_4 + 5e \longrightarrow K^+ + Mn^{2+} + 4O^{-2}$ —— (i)

$H_2S - 2e \longrightarrow 2H^+ + S^\circ$ —— (ii)

(i)×2 and (ii)×5 Then add,

$2KMnO_4 + 10e \rightarrow 2K^+ + 2Mn^{2+} + 8O^{-2}$

$5H_2S - 10e \rightarrow 10H^+ + 5S$

$2KMnO_4 + 5H_2S \rightarrow 5S + 2K^+ + 2Mn^{2+} + 8O^{2-} + 10H^+$

$\Rightarrow 2KMnO_4 + 5H_2S + 6H^+ \rightarrow 5S + 2K^+ + 2Mn^{2+} + 8H_2O$

$\Rightarrow 2KMnO_4 + 5H_2S + 3H_2SO_4 \rightarrow 5S + K_2SO_4 + 2MnSO_4 + 8H_2O$

Ex: $HNO_3 + e \longrightarrow NO_2 + H^+ + O^{-2}$ —— (i)

$CuS - 2e \rightarrow S + Cu^{2+}$ —— (ii)

(i)×2 and (i+ii)

$2HNO_3 + 2e \rightarrow 2NO_2 + 2H^+ + 2O^{-2}$

$CuS - 2e \rightarrow S + Cu^{2+}$

$2HNO_3 + CuS \rightarrow S + 2NO_2 + 2H^+ + 2O^{-2} + Cu^{2+}$

$\Rightarrow 8HNO_3 + CuS \rightarrow 8NO_2 + 4H_2O + CuSO_4$

Ex: $2Na_2S_2O_3 - 2e \longrightarrow Na_2S_4O_6 + 2Na^+$

$I_2 + 2e^- \longrightarrow 2I^-$

$2Na_2S_2O_3 + I_2 \longrightarrow Na_2S_4O_6 + 2NaI$